Grade breakdown

The following criteria are used to mark your Preliminary Design Review. Comments will also be included.

Reporting template	Reporting template guidelines minimally followed and/or over page limit (20 pages plus unlimited appendices). ***0 points***	Template guidelines mostly followed. ***1 points***	All template guidelines followed. ***2 points***
Team organisation and roles	Key project responsibilities not defined. ***0 points***	Key project responsibilities defined and assigned to a team member. ***1 points***	Demonstration of an understanding of what is covered by each role assigned. ***2 points***

Mission overview - Primary Mission	Primary mission not clearly stated, or does not align with that given by CanSat guidelines. ***0 points***	Primary mission clearly defined as per CanSat guidelines. Brief overview of how the primary mission will be achieved (e.g. 'we will use a pressure/temperature sensor'. No specific mention of the use of a ground station/radio for primary mission. ***1 points***	Complete understanding of the primary mission (inclusion of radio / ground station). ***2 points***	Possible hardware and software solutions identified. ***3 points***

Mission overview - Secondary Mission	No suggestions or early indications of secondary mission present and / or secondary mission is prohibited in guidelines (motorised descent). ***0 points***	Some early indications on possible plans for secondary mission. ***1 points***	Basic level of scientific and / or technical reasoning behind the selection / possible selection of the secondary mission. Some thought given to the scientific and / or technical reasoning for the secondary mission / possible secondary missions. ***2 points***	Secondary mission clearl for scientific and / or tecl reasoning, including a we hypothesis and/or 'why?' hardware and software s identified. ***3 points***
Project planning - project plan	No use of a suitable scheduling chart / plan. ***0 points***	Brief overview of a project plan covering the key project areas (i.e. design, programming, construction). ***1 points***	Gantt chart or other suitable tool used to give brief information on project development, with an indication of time required per step. ***2 points***	Gantt chart or other suitable tool used to give detailed information on project development, covering all aspects of the project. ***3 points***

Project planning - team competencies	Unrealistic expectations on scheduling or oversight of a project area/s that could lead to critical project issues. *0 points*	Team competencies mapped to key project areas. *1 points*	Gaps in team competencies and potential support required identified. *2 points*	Thought given to how the team will receive the external support required. *3 points*
Project planning - risk analysis	Risk analysis not present. *0 points*	Some thought given to risk analysis and critical risks identified. *1 points*	Good understanding of the range of risks and consequences across all project areas. *2 points*	Thought given to potential risk mitigations. *3 points*
Mechanical design - CanSat description/diagrams	No high-level description or sketch of the CanSat. *0 points*		High-level description of CanSat mechanical design present, including an early sketch. *1 points*	
Mechanical design - materials	Possible materials not identified. *0 points*	Possible materials identified. *1 points*	Possible materials and their use justified. *2 points*	

Electrical design	Suitable microcontroller / processor not identified. No discussion on possible sensors used to achieve primary / secondary mission. ***0 points***	Suitable microcontroller / processor identified & possible sensors shortlisted. ***1 points***	Microcontroller / processor choice justified. ***2 points***	Good understanding of the capabilities and limitations of the potential sensors identified & compatibility with chosen microcontroller / processor. ***3 points***
Software design	Major software approaches not identified (i.e. which languages / compatibility with hardware). ***0 points***	Overarching approach identified i.e., which software / languages. ***1 points***	Basic block-level diagram included for primary mission only. ***2 points***	Basic block-level diagram for both primary and secondary mission included. ***3 points***
Ground support equipment	Equipment needed not identified or does not meet minimum requirements of primary mission. ***0 points***	Basic overview of equipment needed provided and meets minimum requirements of primary mission. ***1 points***	Choice of equipment justified. ***2 points***	
Testing				

	No test plan included. ***0 points***	Initial test plan included covering some key CanSat guidelines. ***1 points***	Initial test plan included covering all key CanSat guidelines. ***2 points***	Initial test plan included covering all key CanSat guidelines and possible consequences discussed. ***3 points***
Outreach	No outreach plan present. ***0 points***	Basic outreach plan included. ***1 points***	Basic outreach plan and justification of audiences and methods chosen included. ***2 points***	Varied approach to the methods and audiences, outreach plan is creative and engaging. ***3 points***